

EXERCISE SHEET 3

PARTIAL DERIVATIVES

(1) Find domain and range of the following functions

(a) $f(x,y) = \ln(x+y-1)$

(b) $f(x,y) = x^2 e^{3xy}$

(c) $f(x,y) = \sqrt{1+x-y^2}$

(d) $f(x,y,z) = e^{\sqrt{z-x^2-y^2}}$

(a) $\ln$ function is only defined for $x+y-1 > 0$, so $x+y > 1$ that is $y > 1-x$.

$$\text{Domain} = \{ (x,y) \mid y > 1-x \}$$

$$\text{Range} = \mathbb{R}$$

(b) Domain = $\mathbb{R}^2$, Range = $[0, \infty)$ $\because e^{3xy} > 0$ but 0 comes from x^2 , x can be 0. Therefore, Range = $[0, \infty)$

(c) $f(x,y) = \sqrt{1+x-y^2}$ $1+x-y^2 > 0$ $x \geq y^2 - 1$

$$\text{Domain} = \{ (x,y) \mid x \geq y^2 - 1 \}$$
 Range = $[0, \infty)$

(d) $f(x,y,z) = e^{\sqrt{z-x^2-y^2}}$

$$z - x^2 - y^2 \geq 0 \quad z \geq x^2 + y^2$$

Domain = $\{ (x,y,z) \mid z \geq x^2 + y^2 \}$ since exponential function is always positive and the minimum value

of $\sqrt{z-x^2-y^2}$ is 0. Then we get

$$\text{Range} = [1, \infty).$$

② Find the limit, if it exists, or show that the limit does not exist.

(a) $\lim_{(x,y) \rightarrow (6,3)} xy \cos(x-2y)$ (b) $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^4+y^2}$

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

(d) $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xy + yz^2 + xz^2}{x^2 + y^2 + z^2}$

(e) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2}$

(a) $\lim_{(x,y) \rightarrow (6,3)} xy \cos(x-2y) = 6 \cdot 3 \cdot \cos(6-2 \cdot 3) = 18 \cdot 1 = 18$

(b) - Approach (0,0) via $x=0$ $f(0,y) = \frac{0}{y^2} = 0$

$\lim_{\substack{y \rightarrow 0 \\ x=0}} f(0,y) = 0$

- Approach (0,0) via $y=0$ $f(x,0) = \frac{0}{x^4} = 0$

$\lim_{\substack{x \rightarrow 0 \\ y=0}} f(x,0) = 0$

It doesn't mean limit exists. Try $y=x^2$

$f(x,x^2) = \frac{2x^2 \cdot x^2}{x^4 + x^4} = 1$ $\lim_{\substack{x \rightarrow 0 \\ y=x^2}} f(x,x^2) = 1$

For $x=0$ we have $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

For $y=x^2$ we have $\lim_{\substack{x \rightarrow 0 \\ y=x^2}} f(x,x^2) = 1$

$0 \neq 1$ limit doesn't exist.

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$$\textcircled{c} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1} - 1} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2+y^2)(\sqrt{x^2+y^2+1} + 1)}{(x^2+y^2+1) - 1}$$

$$(\sqrt{x^2+y^2+1} + 1) \dots$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{(x^2+y^2)(\sqrt{x^2+y^2+1} + 1)}{(x^2+y^2)} = 2$$

$$\textcircled{d} \quad \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xy + yz^2 + xz^2}{x^2 + y^2 + z^2}$$

- Approach $(0,0,0)$ along x axis, $\Rightarrow y=0 \quad z=0$

$$f(x,0,0) = 0 \quad \lim_{x \rightarrow 0} f(x,0,0) = 0$$

$$y=0 \quad z=0$$

- Approach $(0,0,0)$ along $y=x \quad z=0 \quad f(x,x,0) = \frac{x^2}{2x^2} = \frac{1}{2}$

$$\lim_{\substack{x \rightarrow 0 \\ y=x \\ z=0}} f(x,x,0) = \frac{1}{2} \quad \frac{1}{2} \neq 0 \quad \text{limit doesn't exist.}$$

$$\textcircled{e} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2}$$

Sandwich theorem

$$0 \leq \frac{x^2 \sin^2 y}{x^2 + 2y^2} \leq \sin^2 y$$

Because $x^2 \leq x^2 + 2y^2$

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0$$

$$\frac{x^2}{x^2 + 2y^2} \leq 1$$

$$\lim_{(x,y) \rightarrow (0,0)} \sin^2 y = 0$$

Therefore, S.T. gives

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2} = 0$$

Deyu Attentor

3) $f(x,y) = \begin{cases} \frac{xy}{x^2+xy+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ Is f continuous at $(0,0)$?

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+xy+y^2} = ?$

Approach $(x,y) \rightarrow (0,0)$ along $y=mx$ $f(x,mx) = \frac{mx^2}{x^2+mx^2+m^2x^2}$

$f(x,mx) = \frac{m}{1+m+m^2}$ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{x \rightarrow 0} f(x,mx) = \frac{m}{1+m+m^2}$

For each m , we have different limit values. Therefore, limit doesn't exist. So, it is not continuous at $(0,0)$.

4) Find first partial derivatives of the following functions.

a) $f(x,y) = x^5 + 3x^3y^2 + 3xy^4$

b) $f(x,y) = \frac{x-y}{x+y}$

c) $f(x,y) = x^y$

d) $u = \cos \alpha \cos \beta$

e) $f(x,t) = \arctan(x\sqrt{t})$

f) $u = x e^{-t} \sin \theta$

a) $\frac{\partial f}{\partial x} = f_x = 5x^4 + 9x^2y^2 + 3y^4$ $f_y = 6x^3y + 12xy^3$

b) $f_x = \frac{\partial f}{\partial x} = \frac{(x+y) - (x-y)}{(x+y)^2} = \frac{2y}{(x+y)^2}$

$f_y = \frac{\partial f}{\partial y} = \frac{-(x+y) - (x-y)}{(x+y)^2} = -\frac{2x}{(x+y)^2}$

c) $f_x = y \cdot x^{y-1}$ $f_y = x^y \cdot \ln x$ ($f(x) = a^x$ $f'(x) = a^x \cdot \ln a$)

d) $u_\alpha = \cos \alpha \cos \beta$ $u_\beta = -\sin \beta \cos \alpha$

e) $f_x = \frac{\sqrt{t}}{1+(x\sqrt{t})^2}$ $f_t = \frac{\frac{x}{2\sqrt{t}}}{1+(x\sqrt{t})^2}$

$(\arctan x)' = \frac{1}{1+x^2}$
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④ $u = x e^{-t} \sin \theta$
 $u_x = e^{-t} \sin \theta$ $u_t = -x e^{-t} \sin \theta$ $u_\theta = x e^{-t} \cos \theta$

⑤ Use definition of partial derivatives to obtain f_x, f_y

$$f(x, y) = x^2 - xy + 2y^2$$

$$f_x = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - (x+h)y + 2y^2 - x^2 + xy - 2y^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2hx + h^2 - \cancel{xy} - hy + 2y^2 - \cancel{x^2} + \cancel{xy} - 2y^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2 - hy}{h} = \lim_{h \rightarrow 0} 2x + h - y = 2x - y$$

$$f_y = \frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k} = \lim_{k \rightarrow 0} \frac{x^2 - x(y+k) + 2(y+k)^2 - x^2 + xy - 2y^2}{k}$$

$$= \lim_{k \rightarrow 0} \frac{\cancel{x^2} - \cancel{xy} - xk + 2y^2 + 4ky + 2k^2 - \cancel{x^2} + \cancel{xy} - 2y^2}{k} = \lim_{k \rightarrow 0} \frac{-xk + 4k^2 + 2k^2}{k} = \lim_{k \rightarrow 0} -x + 4k + 2k = -x + 4$$

⑥ $f(x, y) = 4 - x^2 - 2y^2$ obtain $f_y(1, 1)$ using the definition of derivative.

$$f_y(1, 1) = \lim_{k \rightarrow 0} \frac{f(1, 1+k) - f(1, 1)}{k} = \lim_{k \rightarrow 0} \frac{4 - 1 - 2(1+k)^2 - 4 + 1 - 2}{k}$$

$$f(1, 1) = 4 - 1 - 2 = 1$$

$$= \lim_{k \rightarrow 0} \frac{2 - 2(1+k)^2}{k} = \lim_{k \rightarrow 0} \frac{2 - 2 - 4k - 2k^2}{k} = \lim_{k \rightarrow 0} \frac{-4k - 2k^2}{k} = \lim_{k \rightarrow 0} -4 - 2k = -4$$

⑦ Use the definition of partial derivatives for $f(x, y) = \sqrt{3x - y}$
 $f_x(x, y) = ?$ $f_y(x, y) = ?$

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{3(x+h) - y} - \sqrt{3x - y}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{3(x+h) - y} - \sqrt{3x - y}}{h} = \lim_{h \rightarrow 0} \frac{3(x+h) - y - (3x - y)}{h(\sqrt{3(x+h) - y} + \sqrt{3x - y})} = \lim_{h \rightarrow 0} \frac{3h}{h(\sqrt{3(x+h) - y} + \sqrt{3x - y})} = \lim_{h \rightarrow 0} \frac{3}{\sqrt{3(x+h) - y} + \sqrt{3x - y}} = \frac{3}{2\sqrt{3x - y}}$$

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$$f_y(x,y) = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k} = \lim_{k \rightarrow 0} \frac{\sqrt{3x - (y+k)} - \sqrt{3x - y}}{k}$$

$$= \lim_{k \rightarrow 0} \frac{\sqrt{3x - (y+k)} - \sqrt{3x - y}}{k} \cdot \frac{\sqrt{3x - (y+k)} + \sqrt{3x - y}}{\sqrt{3x - (y+k)} + \sqrt{3x - y}} =$$

$$= \lim_{k \rightarrow 0} \frac{3x - (y+k) - (3x - y)}{k (\sqrt{3x - (y+k)} + \sqrt{3x - y})} = \lim_{k \rightarrow 0} \frac{\cancel{3x} - \cancel{y} - k - \cancel{3x} + \cancel{y}}{k (\sqrt{3x - (y+k)} + \sqrt{3x - y})}$$

$$= \lim_{k \rightarrow 0} \frac{-1}{(\sqrt{3x - (y+k)} + \sqrt{3x - y})} = \frac{-1}{(\sqrt{3x - y} + \sqrt{3x - y})} = \frac{-1}{2\sqrt{3x - y}}$$

8 Obtain $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ if $z = z(x, y)$, for the following functions

(a) $x^2 + y^2 + z^2 = 3xyz$

(b) $yz = \ln(x+z)$

(a) $F(x, y, z) = x^2 + y^2 + z^2 - 3xyz = 0$

$$\frac{\partial F}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial F}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial F}{\partial z} \cdot \frac{dz}{dx} = 0 \Rightarrow \frac{dz}{dx} = - \frac{F_x}{F_z}$$

$$\frac{dz}{dx} = - \frac{(2x - 3yz)}{2z - 3xy}$$

$$\frac{dz}{dy} = - \frac{F_y}{F_z} = - \frac{2y - 3xz}{2z - 3xy}$$

(b) $F(x, y, z) = yz - \ln(x+z) = 0$

$$\frac{dz}{dx} = - \frac{F_x}{F_z} = \frac{\frac{1}{x+z}}{y - \frac{1}{x+z}} = \frac{1}{xy + yz - 1}$$

$$\frac{dz}{dy} = - \frac{F_y}{F_z} = \frac{z}{y - \frac{1}{x+z}} = \frac{-z \cdot (x+z)}{xy + xz - 1}$$

(6)

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9 Obtain z_x, z_y for the following functions.

a) $z = f(x) \cdot g(y)$

b) $z = f(x/y)$

a) $z_x = f'(x) \cdot g(y)$ $z_y = g'(y) \cdot f(x)$

b) $x/y = t$ $z = f(t)$ $\frac{\partial z}{\partial x} = \frac{dz}{dt} \cdot \frac{\partial t}{\partial x}$ $\frac{\partial z}{\partial y} = \frac{dz}{dt} \cdot \frac{\partial t}{\partial y}$

$$z_x = f'(t) \cdot \frac{1}{y} = f'(x/y) \cdot \frac{1}{y}$$

$$z_y = f'(t) \cdot \left(-\frac{x}{y^2}\right) = f'(x/y) \cdot \left(-\frac{x}{y^2}\right)$$

10 Show that $u_{xy} = u_{yx}$ for the following functions.

a) $u = x \cdot \sin(x+2y)$

b) $u = x^4 y^2 - 2x y^5$

a) $u_x = \sin(x+2y) + \cos(x+2y) \cdot x$

$$u_{xy} = 2 \cos(x+2y) - 2 \sin(x+2y) \cdot x$$

$$u_{xy} = u_{yx}$$

$$u_y = 2x \cos(x+2y)$$

$$u_{yx} = 2 \cos(x+2y) - 2x \cdot \sin(x+2y)$$

b) $u_x = 4x^3 y^2 - 2y^5$

$$u_{xy} = 8x^3 y - 10y^4$$

$$u_y = 2x^4 y - 10x y^4$$

$$u_{yx} = 8x^3 y - 10y^4$$

$$\boxed{u_{xy} = u_{yx}}$$

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11 Find the following partial derivatives.

a) $f(x,y) = 3xy^4 + x^3y^2$ $f_{xx}y, f_{yxy}$

b) $f(x,t) = x^2 e^{-ct}$ f_{ttt}, f_{txx}

a) $f_x = 3y^4 + 3x^2y^2$

$f_{xx} = 6xy^2$

$f_{xxy} = 12xy$

$f_y = 12xy^3 + 2x^3y$

$f_{yy} = 36xy^2 + 2x^3$

$f_{yyy} = 72xy + 2x^3$

b) $f_t = -cx^2 e^{-ct}$

$f_{tt} = c^2 x^2 e^{-ct}$

$f_{ttt} = -c^3 x^2 e^{-ct}$

$f_{tx} = -2cx e^{-ct}$

$f_{txx} = -2c e^{-ct}$

12 Find equation of the tangent planes for the following surfaces at given points.

a) $z = 4x^2 - y^2 + 2y$ $(-1, 2, 4)$

b) $z = \sqrt{4 - x^2 - 2y^2}$ $(1, -1, 1)$

c) $z = y \ln x$ $(1, 4, 0)$

a) $f_x = 8x$ $f_y = -2y + 2$ $f_x(-1, 2, 4) = -8$

$f_y(-1, 2, 4) = -4 + 2 = -2$

Equation of tangent plane at point $(-1, 2, 4)$

$z - 4 = -8(x - (-1)) - 2(y - 2)$

$z = -8x - 2y$

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$$(b) \quad z_x = \frac{-2x}{2\sqrt{4-x^2-2y^2}} = \frac{-x}{\sqrt{4-x^2-2y^2}} \quad z_x|_{(1,-1,1)} = \frac{-1}{1} = -1$$

$$z_y = \frac{-4y}{2\sqrt{4-x^2-2y^2}} = \frac{-2y}{\sqrt{4-x^2-2y^2}} \quad z_y|_{(1,-1,1)} = 2$$

Report plane at $(1, -1, 1)$

$$z - 1 = -1(x - 1) + 2(y - (-1))$$

$$x - 2y + z = 4$$

$$(c) \quad z_x = \frac{y}{x} \quad z_x|_{(1,4,0)} = 4$$

$$z_y = \ln x \quad z_y|_{(1,4,0)} = \ln 1 = 0$$

Point $(1, 4, 0)$

$$z - 0 = 4(x - 1) + 0(y - 4)$$

$$z = 4x - 4$$

(13) Find linear approximation $L(x, y)$ of the function.

$f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$ and use it to approximate $f(1.95, 1.08)$

$$L(x, y) = f(2, 1) + f_x(2, 1)(x - 2) + f_y(2, 1)(y - 1)$$

$$f(2, 1) = \sqrt{20 - 4 - 7} = 3 \quad f_x = \frac{-2x}{2\sqrt{20 - x^2 - 7y^2}} \quad f_x(2, 1) = \frac{-4}{2 \cdot 3} = -\frac{2}{3}$$

$$f_y(2, 1) = \frac{-14y}{2\sqrt{20 - x^2 - 7y^2}} = -\frac{7}{3}$$

$$L(x, y) = 3 - \frac{2}{3}(x - 2) - \frac{7}{3}(y - 1)$$

$$L(1.95, 1.08) = 3 - \frac{2}{3}(-0.05) - \frac{7}{3}(0.08) = 2.84\bar{6}$$

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14 Find the linear approximation of the function

$f(x,y,z) = \sqrt{x^2+y^2+z^2}$ at $(3,2,6)$ and use it to approximate the number $\sqrt{(3.02)^2 + (1.97)^2 + (5.99)^2}$

$$f_x = \frac{2x}{2\sqrt{x^2+y^2+z^2}} \quad f_y = \frac{2y}{2\sqrt{x^2+y^2+z^2}} \quad f_z = \frac{2z}{2\sqrt{x^2+y^2+z^2}}$$

$$f_x|_{(3,2,6)} = \frac{3}{\sqrt{9+4+36}} = \frac{3}{7} \quad f_y|_{(3,2,6)} = \frac{2}{7} \quad f_z|_{(3,2,6)} = \frac{6}{7}$$

$$f|_{(3,2,6)} = 7 \quad L(x,y,z) = 7 + \frac{3}{7}(x-3) + \frac{2}{7}(y-2) + \frac{6}{7}(z-6)$$

$$L(3.02, 1.97, 5.99) = 7 + \frac{3}{7}(0.02) + \frac{2}{7}(-0.03) + \frac{6}{7}(-0.01)$$

$$\sqrt{(3.02)^2 + (1.97)^2 + (5.99)^2} \approx 6.9914$$

15 Find total differential of the following functions

a) $z = x^3 \ln(y^2)$

b) $u = e^t \sin \theta$

c) $u = \frac{r}{s+2t}$

a) $dz = z_x dx + z_y dy = 3x^2 \ln(y^2) dx + x^3 \cdot \frac{2y}{y^2} dy$

b) $du = z_t dt + z_\theta d\theta = e^t \sin \theta dt + \cos \theta \cdot e^t d\theta$

c) $du = u_r dr + u_s ds + u_t dt$

$$= \frac{1}{s+2t} dr - \frac{r}{(s+2t)^2} ds - \frac{2r}{(s+2t)^2} dt$$

(16) The dimensions of a rectangular box are measured to be 75 cm, 60 cm, and 40 cm, and each measurement is correct to within 0.2 cm. Use differentials to estimate the largest possible error when the volume of the box is calculated from these measurements.

$$V = x \cdot y \cdot z$$

$$dV = yz dx + xz dy + xy dz$$

We are given that $| \Delta x | \leq 0.2$, $| \Delta y | \leq 0.2$, $| \Delta z | \leq 0.2$.

To find the largest error, we take $dx = 0.2$, $dy = 0.2$, $dz = 0.2$.

$$\Delta V \approx dV = 75 \cdot 60 \cdot 0.2 + 60 \cdot 40 \cdot 0.2 + 75 \cdot 40 \cdot 0.2 \\ = 1980$$

Thus, the error just $\Delta \leq 0.2$ for each dimension gives 1980 error in volume.

(17) If $z = x^2 - xy + 3y^2$ and (x, y) changes from $(3, -1)$ to $(2.96, -0.95)$ compute values of Δz and dz .

$$\Delta x = \frac{x}{2.96} - \frac{9}{3} = -0.04$$

$$(a, b) = (3, -1)$$

$$(x, y) = (2.96, -0.95)$$

$$\Delta y = -0.95 - (-1) = 0.05$$

$$f_x = 2x - y \quad f_x(3, -1) = 7$$

$$f_y = -x + 6y \quad f_y(3, -1) = -9$$

$$dz = f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)$$

$$dz = 7(-0.04) - 9(0.05) = -0.73$$

$$\Delta z = (2.96)^2 - (2.96)(-0.95) + 3(0.95)^2 = -0.7189$$

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18 Use chain rules for following functions

(a) $z = x^2y + xy^2$ $x = 2 + t^4$ $y = 1 - t^3$ $dz/dt = ?$

(b) $z = \arctan(2x + y)$ $x = s^2t$ $y = s \ln t$ $\frac{\partial z}{\partial s} = ?$ $\frac{\partial z}{\partial t} = ?$

(c) $w = xy + yz^2$ $x = e^t$ $y = e^t \sin t$ $z = e^t \cos t$ $\frac{dw}{dt} = ?$

(d) $z = \sin \alpha + \tan \beta$ $\alpha = 3s + t$ $\beta = s - t$ $\frac{\partial z}{\partial s} = ?$ $\frac{\partial z}{\partial t} = ?$

(e) $u = \sqrt{r^2 + s^2}$ $r = y + x \cos t$ $s = x + y \sin t$

Obtain $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial t}$ when $x=1$, $y=2$, $t=0$

(f) $R = \ln(u^2 + v^2 + w^2)$ $u = x + 2y$ $v = 2x - y$ $w = 2xy$

Compute $\frac{\partial R}{\partial x}$, $\frac{\partial R}{\partial y}$ when $x=y=1$.

19 $z = f(x, y)$ $x = g(t)$ $y = h(t)$ $g(3) = 2$, $g'(3) = 5$, $h(3) = 7$, $h'(3) = 4$
 $f_x(2, 7) = 6$ $f_y(2, 7) = -8$ $\frac{dz}{dt} \Big|_{t=3} = ?$

20 If $v = f(p, q, r)$ $p = p(x, y, t)$ $q = q(x, y, t)$ $r = r(x, y, t)$
 write $\frac{\partial v}{\partial x} = ?$

21 Obtain (a) $\frac{dy}{dx}$ for $y = y(x)$ $\sqrt{xy} = 1 + x^2y$
 (b) $\frac{dy}{dx}$ for $y = y(x)$ $\cos(x-y) = xe^y$

22 Obtain $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the following equations.

(a) $x - z = \arctan(yz)$

(b) $yz = \ln(x+z)$

(17)

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$$a) \frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= (2xy + y^2) \cdot (4t^3) + (x^2 + 2xy) (-3t^2)$$

$$b) \frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} = \frac{2}{1+(2x+y)^2} \cdot 2st + \frac{1}{1+(2x+y)^2} \ln t$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t} = \frac{2}{1+(2x+y)^2} s^2 + \frac{1}{1+(2x+y)^2} \frac{s}{t}$$

$$c) \frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$= y \cdot e^t + (x+z^2)(e^t \sin t + \cos t e^t) + 2zy \cdot (e^t \cos t - \sin t e^t)$$

$$d) \frac{\partial z}{\partial s} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial s} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial s} = \cos \alpha \cdot \tan \beta \cdot 3 + \sin \alpha (1 + \tan^2 \beta) \cdot 1$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial t} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial t} = \cos \alpha \tan \beta \cdot 1 + (1 + \tan^2 \beta) \cdot \sin \alpha \cdot (-1)$$

$$e) \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} = \frac{2r}{\sqrt{r^2+s^2}} \cdot \cos t + \frac{2s}{\sqrt{r^2+s^2}} \cdot 1$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} = \frac{2r}{\sqrt{r^2+s^2}} \cdot 1 + \frac{s}{\sqrt{r^2+s^2}} \cdot \sin t$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial t} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial t} = \frac{r}{\sqrt{r^2+s^2}} (-x \sin t) + \frac{s}{\sqrt{r^2+s^2}} (y \cos t)$$

For $x=1$, $y=2$, $t=0$ we have $s = x+y \sin t = 1$ (r, s) = (1, 1)
 $r = y+x \cos t = 3$

$$\left. \frac{\partial u}{\partial x} \right|_{\substack{r=3 \\ s=1 \\ x=1 \\ y=2 \\ t=0}} = \frac{4}{\sqrt{10}}$$

$$\left. \frac{\partial u}{\partial y} \right|_{\substack{r=3 \\ s=1 \\ x=1 \\ y=2 \\ t=0}} = \frac{3}{\sqrt{10}}$$

$$\left. \frac{\partial u}{\partial t} \right|_{\substack{r=3 \\ s=1 \\ x=1 \\ y=2 \\ t=0}} = \frac{2}{\sqrt{10}}$$

$$\textcircled{f} \quad \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial z}{\partial w} \cdot \frac{\partial w}{\partial x}$$

$$= \frac{2u}{u^2+v^2+w^2} \cdot 1 + \frac{2v}{u^2+v^2+w^2} \cdot 2 + \frac{2w}{u^2+v^2+w^2} \cdot 2y$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial z}{\partial w} \cdot \frac{\partial w}{\partial y}$$

$$= \frac{2u}{u^2+v^2+w^2} \cdot 2 + \frac{2v}{u^2+v^2+w^2} \cdot (-1) + \frac{2w}{u^2+v^2+w^2} \cdot 2x$$

$$x=y=1 \Rightarrow u=3 \quad v=1 \quad w=2.$$

$$\left. \frac{\partial z}{\partial x} \right|_{\substack{x=y=1 \\ u=3 \\ v=1 \\ w=2}} = \frac{9}{7} \quad \left. \frac{\partial z}{\partial y} \right|_{\substack{x=y=1 \\ u=3 \\ v=1 \\ w=2}} = \frac{9}{7}$$

$$\textcircled{19} \quad \frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} = z_x \cdot g'(t) + z_y \cdot h'(t)$$

$$\left. z_x \right|_{t=3} = f_x(2,7) = 6.$$

$$\text{For } t=3 \quad x=g(3)=2 \quad y=h(3)=7$$

$$\left. z_y \right|_{t=3} = f_y(2,7) = -8.$$

$$\left. \frac{dz}{dt} \right|_{t=3} = \underbrace{z_x \Big|_{t=3}}_6 \cdot \underbrace{g'(3)}_5 + \underbrace{z_y \Big|_{t=3}}_{-8} \cdot \underbrace{h'(3)}_{-4} = 62.$$

$$\textcircled{20} \quad \frac{\partial v}{\partial x} = \frac{\partial v}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial v}{\partial q} \cdot \frac{\partial q}{\partial x} + \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial x} = v_p \cdot p_x + v_q \cdot q_x + v_r \cdot r_x$$

$$\textcircled{21} \quad \textcircled{a} \quad \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{\frac{y}{2\sqrt{xy}} - 2xy}{\frac{x}{2\sqrt{xy}} - x^2}$$

$$F(x,y) = \sqrt{xy} + x^2y - 1$$

$$\textcircled{b} \quad F(x,y) = \cos(x-y) - xe^y$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{-\sin(x-y) - e^y}{-\sin(x-y)(-1) - xe^y}$$

Danke Altintan

$$\textcircled{a} \quad \frac{\partial z}{\partial x} = - \frac{F_x}{F_z} = - \frac{1}{-1 - \frac{y}{1+(y+z)^2}}$$

$$F(x,y,z) = x+z - \arctan(yz)$$

$$\frac{\partial z}{\partial y} = - \frac{F_y}{F_z} = - \frac{\frac{z}{1+(yz)^2}}{-1 - \frac{y}{1+(yz)^2}}$$

$$\textcircled{b} \quad F(x,y,z) = yz - \ln(x+z)$$

$$z_x = - \frac{F_x}{F_z} = - \frac{\frac{1}{x+z}}{y - \frac{1}{x+z}}$$

$$z_y = - \frac{F_y}{F_z} = - \frac{z}{y - \frac{1}{x+z}}$$

21 For the following functions

① Find the gradient of f ② Evaluate the gradient at point P

③ Find the rate of change of f at P in the direction of the vector u

① $f(x,y) = 5xy^2 - 4x^3y$ $P(1,2)$ $u = \left\langle \frac{5}{13}, \frac{12}{13} \right\rangle$

② $f(x,y) = y \ln x$ $P(1,-3)$ $u = \left\langle -\frac{4}{5}, \frac{3}{5} \right\rangle$

③ $f(x,y) = x e^{2yz}$ $P(3,0,2)$ $u = \left\langle \frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right\rangle$

④ $\nabla f = \langle f_x, f_y \rangle = \langle 5y^2 - 12x^2y, 10xy - 4x^3 \rangle$

⑤ $\Delta f|_{P(1,2)} = \langle 5 \cdot 2^2 - 12 \cdot (1)^2(2), 10 \cdot (1)(2) - 4(1)^3 \rangle = \langle -4, 16 \rangle$

⑥ $D_u f(1,2) = \nabla f(1,2) \cdot \vec{u} = \langle -4, 16 \rangle \cdot \left\langle \frac{5}{13}, \frac{12}{13} \right\rangle$

$$= -\frac{20}{13} + \frac{16 \cdot 12}{13} = \frac{172}{13}$$

Deyra Attenton

$$(2) f(x,y) = y \ln x \quad (a) \nabla f = \langle f_x, f_y \rangle = \langle \frac{y}{x}, \ln x \rangle$$

$$(b) \nabla f(1,-3) = \langle -\frac{3}{1}, \ln 1 \rangle = \langle -3, 0 \rangle$$

$$(c) D_u f(1,-3) = \nabla f(1,-3) \cdot \vec{u} = \langle -3, 0 \rangle \cdot \langle -\frac{4}{5}, \frac{3}{5} \rangle = \frac{12}{5}$$

$$(3) f(x,y,z) = x e^{2yz} \quad (a) \nabla f = \langle f_x, f_y, f_z \rangle = \langle e^{2yz}, 2xz e^{2yz}, 2xy e^{2yz} \rangle$$

$$(b) \nabla f(3,0,2) = \langle e^0, 2 \cdot 3 \cdot e^0, 0 \rangle = \langle 1, 12, 0 \rangle$$

$$(c) D_u f(3,0,2) = \nabla f(3,0,2) \cdot \vec{u} = \langle 1, 12, 0 \rangle \cdot \langle \frac{2}{3}, \frac{2}{3}, \frac{1}{3} \rangle = \frac{22}{3}$$

(22) Find the directional derivative of the function at the given point in the direction of the vector v .

$$(1) f(x,y) = \ln(x^2 + y^2) \quad (2,1) \quad v = \langle -1, 2 \rangle$$

$$(2) g(s,t) = s^2 e^t \quad (2,0) \quad v = \vec{i} + \vec{j}$$

$$(3) g(r,\theta) = e^{-r} \sin \theta \quad (0, \pi/3) \quad v = 3\vec{i} - 2\vec{j}$$

$$(4) f(x,y,z) = \sqrt{x^2 + y^2 + z^2} \quad (1,2,-2) \quad v = \langle -6, 6, -3 \rangle$$

$$(1) \nabla f = \langle \frac{2x}{x^2+y^2}, \frac{2y}{x^2+y^2} \rangle \quad \nabla f(2,1) = \langle \frac{4}{5}, \frac{2}{5} \rangle$$

$$v = \langle -1, 2 \rangle \text{ write in unit form } u = \frac{\vec{v}}{|\vec{v}|} = \frac{\langle -1, 2 \rangle}{\sqrt{5}}$$

$$D_u f(2,1) = \nabla f(2,1) \cdot u = \langle \frac{4}{5}, \frac{2}{5} \rangle \cdot \langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \rangle = 0$$

$$(2) \nabla g = \langle g_s, g_t \rangle = \langle 2s e^t, s^2 e^t \rangle \quad \nabla g(2,0) = \langle 4, 4 \rangle$$

$$\vec{u} = \vec{i} + \vec{j} = \langle 1, 1 \rangle \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$$

$$D_u g(2,0) = \langle 4, 4 \rangle \cdot \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle = \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

$$\textcircled{3} g(r, \theta) = e^{-r} \sin \theta \quad \nabla g = \langle g_r, g_\theta \rangle = \langle -e^{-r} \sin \theta, e^{-r} \cos \theta \rangle$$

$$\nabla g|_{(0, \pi/3)} = \langle -e^{-0} \sin \pi/3, e^{-0} \cos \pi/3 \rangle = \langle -\frac{\sqrt{3}}{2}, \frac{1}{2} \rangle$$

$$v = 3\hat{i} - 2\hat{j} = \langle 3, -2 \rangle \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{\langle 3, -2 \rangle}{\sqrt{13}} = \langle \frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}} \rangle$$

$$D_{\vec{u}} g(0, \pi/3) = \langle \frac{\sqrt{3}}{2}, \frac{1}{2} \rangle \cdot \langle \frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}} \rangle = \frac{3\sqrt{3}}{2\sqrt{13}} - \frac{2}{2\sqrt{13}} = \frac{3\sqrt{3}-2}{2\sqrt{13}}$$

$$\textcircled{4} f(x, y, z) = \sqrt{x^2 + y^2 + z^2} \quad (1, 2, -2) \quad v = \langle -6, 6, -3 \rangle$$

$$\nabla f = \left\langle \frac{x}{\sqrt{x^2+y^2+z^2}}, \frac{y}{\sqrt{x^2+y^2+z^2}}, \frac{z}{\sqrt{x^2+y^2+z^2}} \right\rangle \quad \nabla f(1, 2, -2) = \left\langle \frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right\rangle$$

$$v = \langle -6, 6, -3 \rangle \quad u = \frac{\vec{v}}{|\vec{v}|} = \frac{\langle -6, 6, -3 \rangle}{\sqrt{36+36+9}} = \left\langle -\frac{6}{9}, \frac{6}{9}, -\frac{3}{9} \right\rangle = \left\langle -\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right\rangle$$

$$D_{\vec{u}} f = \nabla f(1, 2, -2) \cdot \vec{u} = \left\langle \frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right\rangle \cdot \left\langle -\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right\rangle = -\frac{2}{9} + \frac{4}{9} + \frac{2}{9} = \frac{4}{9}$$

23 For the following functions

- (a) Find the maximum rate of change (increases most rapidly)
 (b) " " minimum " " " (decreases most rapidly)

(c) zero change

(1) $f(x, y) = y^2/x \quad (2, 4)$ (2) $f(p, q) = qe^{-p} + pe^{-q} \quad (0, 0)$

(3) $f(x, y, z) = x^2 y z^4 \quad (1, 1, 1)$ (4) $f(x, y, z) = \tan(x+2y+3z) \quad (-5, 1)$

(1) $\nabla f = \left\langle -\frac{y^2}{x^2}, \frac{2y}{x} \right\rangle \quad \nabla f|_{(2, 4)} = \langle -4, 4 \rangle$

(2) $\vec{u} = \frac{\nabla f}{|\nabla f|} = \frac{\langle -4, 4 \rangle}{\sqrt{32}} = \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$ direction of maximum rate of change.

the maximum rate $|\nabla f| = \sqrt{32} = 4\sqrt{2}$

Daya Attinter

(b) Decreases most rapidly in the direction of $-u = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$

minimum rate of change $-\nabla f = -4\sqrt{2}$

(c) zero change at point $(2, 4)$ is $\vec{u}$ which is orthogonal to ∇f

$$\nabla f|_{(2,4)} = \langle -4, 4 \rangle \quad \vec{u} = \langle u_1, u_2 \rangle$$

$$\nabla f|_{(2,4)} \cdot \vec{u} = -4 \cdot u_1 + 4u_2 = 0$$

$$u_1 = u_2$$

Since the length is 1. $\sqrt{u_1^2 + u_2^2} = 1$. $\sqrt{2u_1^2} = 1$

$u_1 = \frac{1}{\sqrt{2}}$ so the zero change at the direction

$$u = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle \quad \text{and} \quad -u = \langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$$

(2) $f(p, q) = qe^{-p} + pe^{-q}$

$$\nabla f(p, q) = \langle -qe^{-p} + e^{-q}, e^{-p} - pe^{-q} \rangle$$

$$\nabla f = \langle f_p, f_q \rangle$$

$$\nabla f(0,0) = \langle 1, 1 \rangle$$

(a) $\vec{u} = \frac{\nabla f}{|\nabla f|} = \frac{\langle 1, 1 \rangle}{\sqrt{2}} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$ direction of max. rate

$$|\nabla f| = \sqrt{2} \text{ max rate}$$

(b) $-\vec{u} = -\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle = \langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$ $-|\nabla f| = -\sqrt{2}$ min rate of change

(c) zero direction $\vec{u} = \langle u_1, u_2 \rangle$ $\nabla f(0,0) \cdot \vec{u} = \langle 1, 1 \rangle \cdot \langle u_1, u_2 \rangle = u_1 + u_2 = 0$

$u_1 = -u_2$ $\sqrt{u_1^2 + u_2^2} = 1$ $\sqrt{2u_1^2} = 1$ Direction of zero change $\vec{u} = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$

(3) $f(x, y, z) = x^2 y^2 z^2$ $(1, 1, 1)$

$$\nabla f = \langle 2xy^2z^2, 2x^2yz^2, 2x^2y^2z \rangle$$

$$\nabla f(1, 1, 1) = \langle 2, 2, 2 \rangle$$

(a) $\vec{u} = \frac{\nabla f}{|\nabla f|} = \frac{\langle 2, 2, 2 \rangle}{\sqrt{12}}$ direction of max rate of change

max rate $\sqrt{12}$

(b)

Deyu Altinter

$$(b) -\vec{u} = -\frac{\nabla f}{|\nabla f|} = -\frac{\langle 2, 2, 2 \rangle}{\sqrt{12}}, \text{ min rate } -\sqrt{12}.$$

$$(c) \text{ zero } \nabla f \cdot \langle u_1, u_2, u_3 \rangle = 2u_1 + 2u_2 + 2u_3 = 0$$

$$\vec{u} = \langle u_1, u_2, u_3 \rangle$$

$$u_1 + u_2 + u_3 = 0$$

$$\sqrt{u_1^2 + u_2^2 + u_3^2} = 0$$

$$\text{Let } u_1 = t, u_2 = s, \text{ then, } u_3 = -s - t.$$

$$\sqrt{t^2 + s^2 + s^2 - 2st + t^2} = 1 \quad \vec{u} = \{ \langle t, s, -s-t \rangle : \text{such that}$$

$$\sqrt{2(t^2 + s^2) - 2st} = 1 \}$$

$$(4) \nabla f = \langle \sec^2(x+2y+3z) \cdot 1, \sec^2(x+2y+3z) \cdot 2, \sec^2(x+2y+3z) \cdot 3 \rangle$$

$$\nabla f(-5, 1, 1) = \langle \sec^2(0), 2\sec^2(0), 3\sec^2(0) \rangle = \langle 1, 2, 3 \rangle$$

$$(a) \vec{u} = \frac{\nabla f}{|\nabla f|} = \frac{\langle 1, 2, 3 \rangle}{\sqrt{14}} = \left\langle \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle \quad \underbrace{|\nabla f| = \sqrt{14}}_{\text{max value}}$$

max direction

$$(b) -\vec{u} = -\frac{\nabla f}{|\nabla f|} = \frac{\langle -1, -2, -3 \rangle}{\sqrt{14}} \quad \underbrace{|\nabla f| = \sqrt{14}}_{\text{min value.}}$$

min direction

$$(c) \vec{u} = \langle u_1, u_2, u_3 \rangle \quad \nabla f \cdot \vec{u} = u_1 + 2u_2 + 3u_3 = 0$$

$$\sqrt{u_1^2 + u_2^2 + u_3^2} = 0$$

$$u_1 = s, u_2 = t, u_3 = \frac{-s-2t}{3}$$

$$u = \left\{ \left\langle s, t, \frac{-s-2t}{3} \right\rangle : \sqrt{s^2 + t^2 + \left(\frac{-s-2t}{3}\right)^2} = 1 \right\}$$

24 Find equations of (a) tangent plane (b) normal line for the given surface at the specified point

(1) $x^2 + 2y^2 + 3z^2 = 21$ $(4, -1, 1)$

(2) $x = y^2 + z^2 - 2$ $(-1, 1, 0)$

(3) $x - z = 4 \text{ or } \tan(4z)$ $(1 + \pi, 1, 1)$

(4) $z + 1 = x e^y \cos z$ $(1, 0, 0)$

(1) $f(x, y, z) = x^2 + 2y^2 + 3z^2$ $\nabla f = \langle 2x, 4y, 6z \rangle$ $\nabla f|_{(4, -1, 1)} = \langle 8, -4, 6 \rangle$

Tangent plane $8(x - 4) - 4(y + 1) + 6(z - 1) = 0$

Normal line $\frac{x - 4}{8} = \frac{y + 1}{-4} = \frac{z - 1}{6}$

(2) $f(x, y, z) = y^2 + z^2 - x$ $\nabla f = \langle -1, 2y, 2z \rangle$ $\nabla f|_{(-1, 1, 0)} = \langle -1, 2, 0 \rangle$

Tangent plane $-1(x + 1) + 2(y - 1) + 0(z - 0) = 0$
 $-x + 2y = 3$

Normal line $\frac{x + 1}{-1} = \frac{y - 1}{2}, z = 0$

(3) $F(x, y, z) = x - z - 4 \text{ or } \tan(4z)$

$\nabla F = \left\langle 1, \frac{-4z}{1 + y^2 z^2}, -1 - \frac{4y}{1 + y^2 z^2} \right\rangle$

$\nabla F|_{(1 + \pi, 1, 1)} = \left\langle 1, -\frac{4}{1 + 1}, -1 - \frac{4}{1 + 1} \right\rangle = \langle 1, -2, -3 \rangle$

Tangent $1(x - (1 + \pi)) - 2(y - 1) - 3(z - 1) = 0$

Normal $\frac{x - (1 + \pi)}{1} = \frac{y - 1}{-2} = \frac{z - 1}{-3}$

(20)

Darya Altunkar

④ $F(x,y,z) = -z + xe^y \cos z$

$\nabla F = \langle e^y \cos z, xe^y \cos z, -1 - xe^y \sin z \rangle$

$\nabla F(1,0,0) = \langle 1, 1, -1 \rangle$

Open plane

$1(x-1) + 1(y-0) - 1(z-0) = 0$

Normal line

$\frac{x-1}{1} = \frac{y-0}{1} = \frac{z-0}{-1}$

$x-1 = y = -z$

25 Find and classify all the critical points of the following functions

① $f(x,y) = 9 - 2x + 4y - x^2 - 4y^2$

② $f(x,y) = x^3 y + 12x^2 - 8y$

③ $f(x,y) = x^4 + y^4 - 4xy + 2$

④ $f(x,y) = e^{4y-x^2-y^2}$

⑤ $f(x,y) = (1+xy)(x+y)$

① $f_x = -2 - 2x$
 $f_y = 4 - 8y$ } solve together $-2 - 2x = 0 \Rightarrow x = -1$ only critical point
 $4 - 8y = 0 \Rightarrow y = 1/2$
 $(-1, 1/2)$

$f_{xx} = -2$ $f_{xy} = 0$ $f_{yy} = -8$

$D = f_{xx}(-1, 1/2) \cdot f_{yy}(-1, 1/2) - (f_{xy}(-1, 1/2))^2 = (-2)(-8) - 0 = 16 > 0$

$f_{xx}(-1, 1/2) = -2 < 0$ local maximum by the second derivative test (S.D.T) Max value of the function $f(-1, 1/2) = 11$

② $f_x = 3x^2 y + 24x$ $f_x = 0 \Rightarrow 12y + 4x = 0 \Rightarrow y = -4$

$f_y = x^3 - 8$

$f_y = 0 \Rightarrow x^3 - 8 = 0 \Rightarrow x = 2$

only critical point $(2, -4)$

$f_{xx} = 6xy + 24$
 $f_{xx}(2, -4) = -24$

$f_{xy} = 3x^2$ $f_{yy} = 0$
 $f_{xy}(2, -4) = 12$ $f_{yy}(2, -4) = 0$

S.D.T $D = f_{xx}(2, -4) \cdot f_{yy}(2, -4) - (f_{xy}(2, -4))^2$
 $= (-24) \cdot 0 - (12)^2 = -144 < 0$

$f_{xx}(2, -4) = -24 < 0$ Saddle point.

③ $f(x,y) = x^4 + y^4 - 4xy + 2$

$$f_x = 4x^3 - 4y \quad f_{xx} = 12x^2$$

$$f_y = 4y^3 - 4x \quad f_{xy} = -4 \quad f_{yy} = 12y^2$$

Find critical point
 $f_x = 0 = f_y = 0$

$$\begin{aligned} 4x^3 - 4y &= 0 & y &= x^3 \\ 4y^3 - 4x &= 0 & x &= y^3 \end{aligned}$$

$$\begin{aligned} x &= x^9 \\ x - x^9 &= 0 \\ x(1 - x^8) &= 0 \\ x &= 0 & x &= 1 & x &= -1 \\ \downarrow & & \downarrow & & \downarrow \\ y &= 0 & y &= 1 & y &= -1 \end{aligned}$$

Critical points $(0,0)$ $(1,1)$ $(-1,-1)$

S.D.T $D = f_{xx}(0,0)f_{yy}(0,0) - (f_{xy}(0,0))^2$
 $= 0 - 16 = -16 < 0$ saddle point.

(1,1)
 $D = f_{xx}(1,1)f_{yy}(1,1) - (f_{xy}(1,1))^2$
 $= 12(12) - 16 = 144 - 16 = 128 > 0$

$f_{xx}(1,1) = 144 > 0$ local min value

(-1,-1) $D = f_{xx}(-1,-1)f_{yy}(-1,-1) - (f_{xy}(-1,-1))^2 = 144 - 16 = 128 > 0$
 $f_{xx}(-1,-1) = 12 > 0$ local min
 $44 - x^2 - y^2$

④ $f(x,y) = e^{44-x^2-y^2}$
 $f_x = -2xe^{44-x^2-y^2}$ $f_y = (4-2y)e^{44-x^2-y^2}$
 $f_{xx} = (4x^2-2)e^{44-x^2-y^2}$ $f_{xy} = -2x(4-2y)e^{44-x^2-y^2}$

$f_{yy} = (4y^2 - 16y + 14)e^{44-x^2-y^2}$

$f_x = f_y = 0 \Rightarrow x=0 \quad y=2$ Only critical point $(0,2)$
 $e^{44-x^2-y^2} \neq 0$

S.D.T

$D = f_{xx}(0,2)f_{yy}(0,2) - (f_{xy}(0,2))^2$
 $= (-2e^4)(-2e^4) - 0^2 = 4e^8 > 0$

$f_{xx}(0,2) = -2e^4 < 0$ $f(0,2) = e^4$ is a local max value

Dejer Altıntaş

③ $f(x,y) = (1+x^2)(x+y) = x+y+x^2y+xy^2$

$f_x = 1+2xy+y^2 \Rightarrow f_{xx} = 2y$

$f_y = 1+x^2+2xy \quad f_{xy} = 2x+2y$

$f_{yy} = 2x$

$f_x = f_y = 0 \quad \cancel{1+2xy+y^2} = \cancel{1+x^2+2xy}$
 $y^2 = x^2 \quad y = \pm x$

If $y=x$ then $f_x = 1+2x^2+x^2 = 0 \quad 1+3x^2 = 0$ No root.

So, we will only focus $x=-1 \quad f_x = 1-2x^2+x^4 = 0$
 $x^2=1 \quad x = \pm 1$

Critical points $(1,-1) (-1,1)$

S.D.T $D = f_{xx}(1,-1) f_{yy}(1,-1) - (f_{xy}(1,-1))^2 = (2)(2) - 0 < 0$

$D = f_{xx}(-1,1) f_{yy}(-1,1) - (f_{xy}(-1,1))^2 = (-2)(-2) - 0 < 0$

$(1,1), (-1,1)$ are saddle points.

26 Find the absolute maximum and minimum values of f on the set D

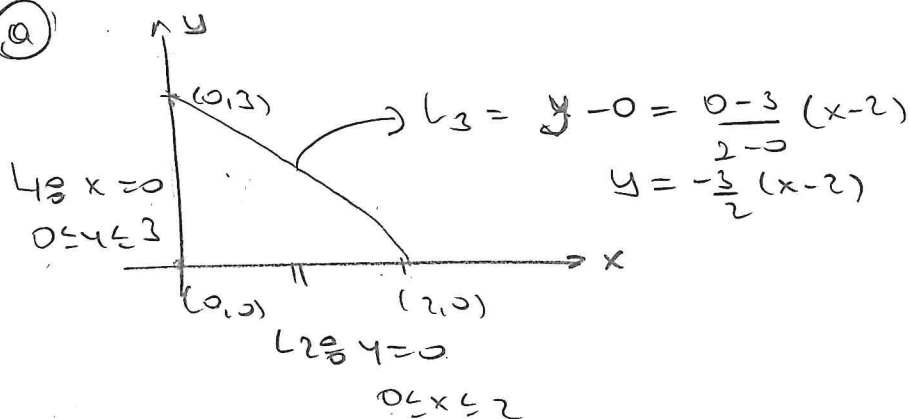
(a) $f(x,y) = 1+4x-5y$ D is the closed triangular region with vertices $(0,0), (2,0)$ and $(0,3)$

(b) $f(x,y) = 3+xy-x-2y$ D is the closed triangular region with vertices $(1,0), (5,0)$ and $(1,4)$

(c) $f(x,y) = x^2+y^2+x^2y+y$ $D = \{(x,y) \mid |x| \leq 1, |y| \leq 1\}$

(d) $f(x,y) = xy^2$ $D = \{(x,y) \mid x \geq 0, y \geq 0, x^2+y^2 \leq 3\}$

(a)



Interior points $f_x = 4 \quad f_y = -5$ No critical point inside the domain.

Boundary parts $L_1: x=0 \quad f(0, y) = 1 - 5y \quad 0 \leq y \leq 3$
 $f'(0, y) = -5$ no point from $f'(0, y) \neq 0$
 Only "initial-final" points $(0, 0) \Rightarrow \boxed{f(0, 0) = 1} \quad \boxed{f(0, 3) = -14}$

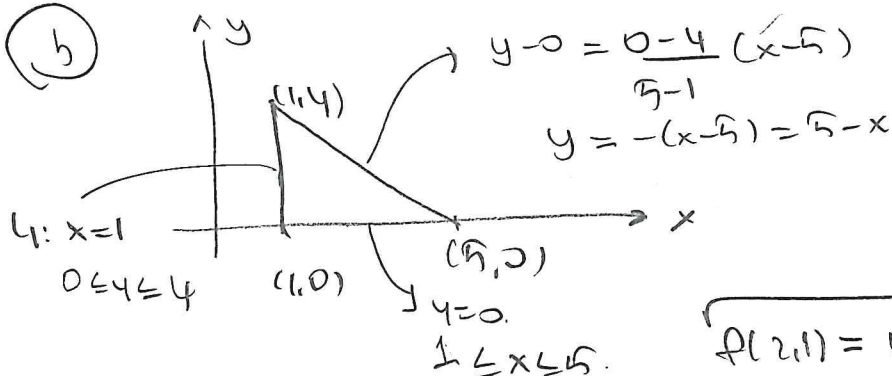
$L_2: y=0 \quad f(x, 0) = 1 + 4x \quad f'(x, 0) = 4 \neq 0$ no point
 Initial-final $\boxed{f(0, 0) = 1} \quad \boxed{f(2, 0) = 9}$

$L_3: y = -\frac{3}{2}(x-2) \quad f(x, -\frac{3}{2}x+3) = \frac{23}{2}x - 14$
 $f'(x, -\frac{3}{2}x+3) = \frac{23}{2} \neq 0$

Initial-final $\boxed{f(2, 0) = 9} \quad \boxed{f(0, 3) = -14}$

Abs. max $f(2, 0) = 9$ Abs. min $f(0, 3) = -14$

(b)



Interior points

$$f_x = y - 1$$

$$f_y = x - 2$$

$$f_x = 0 \quad f_y = 0$$

$$(2, 1)$$

Deja Attinter

$$\boxed{f(2, 1) = 1}$$

$$(2, 4)$$

Boundaries L_1 : $x=1$ $0 \leq y \leq 4$ $f(1,y) = 3+y-1-2y = 2-y$
 $f'(1,y) = -1 \neq 0$ no critical point.

$$\boxed{f(1,0) = 2} \quad \boxed{f(1,4) = -2}$$

L_2 : $y=0$ $1 \leq x \leq 5$ $f(x,0) = 3-x$ $f'(x,0) = -1 \neq 0$

$$\boxed{f(1,0) = 2} \quad \boxed{f(5,0) = -2}$$

L_3 : $y=5-x$ $f(x,5-x) = 3+x(5-x)-x-2(5-x) = -(x-3)^2+2$

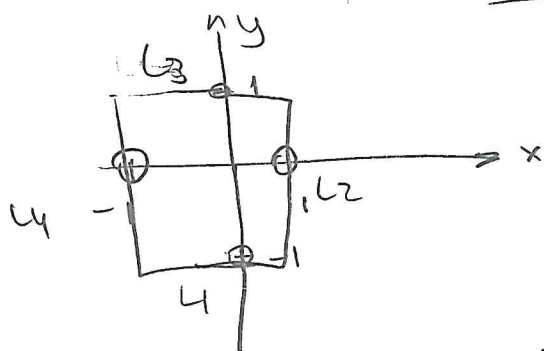
$$f'(x,5-x) = -2(x-3) = 0 \quad x=3 \quad y=5-3=2$$

$$\boxed{f(3,2) = 2} \quad \boxed{f(1,4) = -2} \quad \boxed{f(5,0) = -2}$$

Compare all points absolute max of f is 2 at points $(1,0), (3,2)$

Abs. min of f is -2 at points $(5,0), (1,4)$

(c) $f(x,y) = x^2 + y^2 + x^2y + 4$ $D = \{(x,y) \mid -1 \leq x \leq 1, -1 \leq y \leq 1\}$



Interior point

$$f_x = 2x + 2xy$$

$$f_x = f_y = 0$$

$$f_y = 2y + x^2$$

$$\begin{cases} 2x + 2xy = 0 \\ 2y + x^2 = 0 \end{cases} \quad (0,0)$$

$$\boxed{f(0,0) = 4}$$

L_1 : $y=-1$ $-1 \leq x \leq 1$ $f(x,-1) = x^2 + 1 - x^2 + 4 = 5$ constant
 $f(-1,-1) = f(1,-1) = 5$

L_2 : $x=1$ $-1 \leq y \leq 1$ $f(1,y) = 1 + y^2 + y + 4 = y^2 + y + 5$

$$\boxed{f(1,-1) = 5} \quad \boxed{f(1,0) = 7} \quad \boxed{f(1,-1/2) = 19/4}$$

$$L_3: y=1 \quad -1 \leq x \leq 1 \quad f(x,1) = 2x^2 + 5 \quad f'(x,1) = 4x \quad x=0$$

$$\boxed{f(0,1) = 5}$$

$$L_4: x=-1 \quad -1 \leq y \leq 1 \quad f(-1,y) = y^2 + y + 5$$

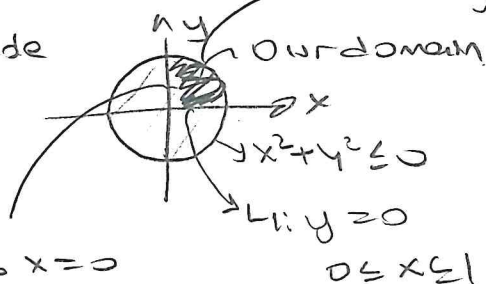
$$f'(-1,y) = 2y + 1 = 0 \quad y = -1/2$$

$$\boxed{f(-1, -1/2) = \frac{19}{4}}$$

$$\boxed{f(-1,1) = 7}$$

Compare all results Abs max 7 at points $(-1,1)(1,1)$
 Abs min $\frac{19}{4}$ at point $(-1, -1/2)$

(d) Interior points $f_x = y^2 \quad f_y = 2xy \quad f_x = 0 = f_y = 0 \quad L_3: x^2 + y^2 = 3$
 $f_x = 0$ for $y=0$ no critical points inside the domain



Boundaries

$$L_1: y=0 \quad 0 \leq x \leq 1 \quad \boxed{f(x,0) = 0}$$

$$L_2: x=0 \quad 0 \leq y \leq 1 \quad \boxed{f(0,y) = 0}$$

$$L_3: x=0 \quad 0 \leq y \leq 1 \quad \boxed{f(0,y) = 0}$$

$$L_5: f(x, \sqrt{3-x^2}) = x \cdot (3-x^2) = 3x - x^3$$

$$f'(x, \sqrt{3-x^2}) = 3 - 3x^2 \quad x = \pm 1 \quad y = \sqrt{2}$$

Critical point $(1, \sqrt{2}) \quad \boxed{f(1, \sqrt{2}) = 2}$

Boundary $\boxed{f(\sqrt{3}, 0) = f(0, \sqrt{3}) = 0}$

Abs max is 2 at point $(1, \sqrt{2})$

absolute min is 0 along all points of L_1 and L_2 .

(27) Find the points on the surface $x^2 y^2 z = 1$ that are closest to the origin

$z = \frac{1}{x^2 y^2}$ Our aim is to minimize $d = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2}$
 $d^2 = x^2 + y^2 + z^2$ use $z = \frac{1}{x^2 y^2}$

$d^2 = x^2 + y^2 + x^{-4} y^{-4}$ $f_x = 2x - \frac{4}{x^5 y^4}$ $f_y = 2y - \frac{4}{x^4 y^5}$

$f_x = 0 \Rightarrow 2x = \frac{4}{x^5 y^4}$, $f_y = 0 \Rightarrow 2y = \frac{4}{x^4 y^5}$
 $\underbrace{x^6 y^4 = 2}$ $\underbrace{x^4 y^6 = 2}$

so $x^6 y^4 = 2 = x^4 y^6$ then $x^2 = y^2 \Rightarrow x = \pm y$

$x^{10} = 2 \Rightarrow x = \pm 2^{1/10}$ $y = \pm 2^{1/10}$

we have 4 critical points $A = (2^{1/10}, 2^{1/10})$ $C = (-2^{1/10}, 2^{1/10})$
 $B = (2^{1/10}, -2^{1/10})$ $D = (-2^{1/10}, -2^{1/10})$

abs min is at that $f(A) = f(B) = f(C) = f(D) = 2^{-2/5}$

No abs max because surface is infinite

(28) Find three positive numbers x, y and z whose sum is 100 such that $x^a y^b z^c$ is a maximum

$x + y + z = 100 \Rightarrow z = 100 - x - y$

$f(x, y) = x^a y^b (100 - x - y)^c$

$f_x = a x^{a-1} y^b (100 - x - y)^c - c x^a y^b (100 - x - y)^{c-1} = x^{a-1} y^b (100 - x - y)^{c-1} [a(100 - x - y) - cy]$

$f_y = x^a y^{b-1} (100 - x - y)^{c-1} [b(100 - x - y) - cx]$

Since x, y, z are all positive.

$f_x = 0$ $f_y = 0$ we have $\left. \begin{aligned} x &= \frac{100b}{a+b+c} \\ y &= \frac{100a}{a+b+c} \end{aligned} \right\}$ use $x + y + z = 100$
 $z = \frac{100c}{a+b+c}$

(27)

Deriva Hunter

(29) Find the volume of the largest rectangular box in the first octant with three faces in the coordinate planes and one vertex in the plane $x+2y+3z=6$

$$\begin{aligned} \text{Maximize } f(x,y,z) &= x \cdot y \cdot z \\ x+2y+3z &= 6 \\ z &= \frac{6-x-2y}{3} \end{aligned} \quad \left\{ \begin{aligned} f(x,y) &= x \cdot y \cdot \left(\frac{6-x-2y}{3} \right) \end{aligned} \right.$$

$$\begin{aligned} f_x &= \frac{1}{3} (6y - 2xy - y^2) = \frac{1}{3} y (6 - 2x - y) & f_x &= 0 \text{ gives critical point.} \\ f_y &= \frac{1}{3} x (6 - x - 4y) & (2,1) \end{aligned}$$

Thus the volume of the largest box is $\sqrt{(2,1,\frac{2}{3})} = 4/3$.

(30) Find the dimensions of the rectangular box with largest volume such that the sum of the lengths of its 12 edges is c

Edges of rectangular box x, y, z

$$f(x,y,z) = 4x + 4y + 4z = c \Rightarrow z = \frac{1}{4} (c - 4x - 4y) = \frac{c}{4} - x - y$$

$$V(x,y,z) = xyz = xy \left(\frac{c}{4} - x - y \right) = \frac{cxy}{4} - x^2y - xy^2$$

$$x > 0 \quad y > 0 \quad V_x = \frac{1}{4}cy - 2xy - y^2 \quad V_y = \frac{1}{4}cx - x^2 - 2xy$$

$$V_x = 0 = V_y \quad \frac{1}{4}c = 2x + y \quad \frac{1}{4}c = x + 2y$$

$$x + 2y = 2x + y$$

$$y = x$$

$$x = \frac{1}{12}c \quad y = \frac{1}{12}c$$

$$z = \frac{1}{4}c - x - y = \frac{1}{12}c$$

$$\sqrt{x=y=z=\frac{1}{12}c}$$

31) Find max-min values of the following functions under given constraints

a) $f(x,y) = x^2 - y^2$ $x^2 + y^2 = 1$ b) $f(x,y) = 4x + 6y$ $x^2 + y^2 = 13$

c) $f(x,y) = x^2 y$ $x^2 + 2y^2 = 6$ d) $f(x,y,z) = 2x + 6y + 10z$

$x^2 + y^2 + z^2 = 35$

e) $f(x,y,z) = 8x - 4z$ $x^2 + 10y^2 + z^2 = 5$

a) $\nabla f = \lambda \nabla g$ $g(x,y) = x^2 + y^2 = 1$

$\langle 2x, -2y \rangle = \lambda \langle 2x, 2y \rangle$

$2x = 2\lambda x$ $-2y = 2\lambda y$

$2x = 2\lambda x \Rightarrow x = 0$ or $x = 1$

$x = 0$ $x^2 + y^2 = 1$ $y^2 = 1$ $y = \pm 1 \Rightarrow (0,1) (0,-1)$ $x^2 + y^2 = 1$

If $\lambda = 1$ then $-2y = 2\lambda y$ implies $y = 0$ so $x = \pm 1$
 $(1,0) (-1,0)$

So, we have $(\pm 1, 0) (0, \pm 1)$

$f(\pm 1, 0) = 1$ $f(0, \pm 1) = -1$
 max value min value

b) $\nabla f = \lambda \nabla g$ $\langle 4, 6 \rangle = \lambda \langle 2x, 2y \rangle$

$g(x,y) = x^2 + y^2 = 13$

$4 = 2\lambda x$ $6 = 2\lambda y$ so $x = \frac{4}{2\lambda}$ $y = \frac{6}{2\lambda}$

Point $(\frac{4}{2\lambda}, \frac{6}{2\lambda})$ Constraint $(\frac{4}{2\lambda})^2 + (\frac{6}{2\lambda})^2 = 13$

$\lambda = \pm 1$

So our points $\lambda = 1$ $(2, 3)$ $f(2,3) = 26$ max
 $\lambda = -1$ $(-2, -3)$ $f(-2,-3) = -26$ min

c) $\nabla f = \lambda \nabla g$ $\langle 2xy, x^2 \rangle = \lambda \langle 2x, 4y \rangle$ $2xy = 2\lambda x$ (I)
 $x^2 = 4\lambda y$ (II)

From (I) we get
 $2xy = 2\lambda x \Rightarrow \underline{x=0}$ or $y=\lambda$

Deja A Hinten

If $\underline{x=0}$, then from (II) $\lambda=0$ or $y=0$
 (I) $\Rightarrow x=0$ (II) $\Rightarrow y=0$ $(0,0)$
 $y=x$

However, if $y=0$ then $g(x,y) = x^2 + 2y^2 = 0$ contradiction

- If $\lambda=0$ then using constraint $g(x,y)=6$ $2y^2=6$ $y=\pm\sqrt{3}$
 $x=0$ $(0, \sqrt{3})$ $(0, -\sqrt{3})$

• If $\underline{y=\lambda}$ then $\downarrow$ $x^2 = 4\lambda y$ we get $x^2 = 4y^2$
 from II

$g(x,y) = 6$ gives $4y^2 + 2y^2 = 6 \Rightarrow y^2 = 1$ $y = \pm 1 \Rightarrow \begin{matrix} x^2 = 4y^2 \\ x = \pm 2 \end{matrix}$

so possible extreme values are at points
 $(0, \sqrt{3}), (0, -\sqrt{3}), (2, 1), (-2, 1), (2, -1), (-2, -1)$

$$f(2, 1) = f(-2, 1) = 4 \quad f(0, \sqrt{3}) = 0$$

$$f(0, -\sqrt{3}) = 0$$

$$f(2, -1) = f(-2, -1) = -4$$

Max 4 at points $(\pm 2, 1)$

Min -4 at points $(\pm 2, -1)$

① $\nabla f = \lambda \nabla g \quad \langle 2, 6, 10 \rangle = \langle 2\lambda x, 2\lambda y, 2\lambda z \rangle$
 $2\lambda x = 2 \Rightarrow x = \frac{1}{\lambda} \quad 6 = 2\lambda y \Rightarrow y = \frac{3}{\lambda} \quad 2\lambda z = 10 \Rightarrow z = \frac{5}{\lambda}$
 Constraint $x^2 + y^2 + z^2 = 35 \quad \frac{1}{\lambda^2} + \frac{9}{\lambda^2} + \frac{25}{\lambda^2} = 35 \quad \lambda = \pm 1$

so possible extreme values at points $(1, 3, 5)$ $(-1, -3, -5)$
 $f(1, 3, 5) = 70$ max, $f(-1, -3, -5) = -70$ min

② $\nabla f = \langle 8, 0, -4 \rangle \quad \nabla f = \lambda \nabla g$
 $\nabla g = \langle 2x, 20y + 2z \rangle \quad \langle 8, 0, -4 \rangle = \lambda \langle 2x, 20y + 2z \rangle$

$$2x\lambda = 8 \quad 20\lambda y = 0 \quad 2\lambda z = -4$$

$$\underline{x = \frac{4}{\lambda}} \quad y = 0 \quad z = -\frac{2}{\lambda}$$

Constraint $x^2 + 10y^2 + z^2 = 5 \quad \left(\frac{4}{\lambda}\right)^2 + 10 \cdot 0^2 + \left(-\frac{2}{\lambda}\right)^2 = 5$

Points $(2, 0, -1), (-2, 0, 1)$ $\lambda = \pm 2$
 ③ $f(2, 0, -1) = 20$ max $f(-2, 0, 1) = -20$ min

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